Lab #3: Arpspoof Warning System

There are multiple ways of detecting ARP spoofing attacks.

Tasks:

- Read up on some of those techniques
  - Go beyond Wikipedia ☺
- Write the Arp spoofing warning tool
  - The warning must go off iff you detect <u>at least</u> two indicators.
- Check that it works against a different computer on LAN (either virtual or real) when running the ARP spoofing tool you wrote in Lab #2.
- Can you initiate a genuine false positive from your tool?

Added challenge (bonus 20% for either challenge):

- Go beyond detection into the protection realm**…** Can you thwart the attack?
  - Think of ideas on how to do this
    - You are allowed to use the internet to get ideas
  - Can you implement them?
- Can you re-write your ARP spoofer (Lab #2) to succeed even when a warning system is active?
  - Take a classmates warning tool and WITHOUT looking at their code try and play with your ARP spoofer to get around their protection

Pointers:

- You will need to programmatically access OS commands (or wrappers to this functionality) to accomplish this lab.
- You will need Administrator privileges to run some functionality (specifically some of the added challenges).
- Look into the `filter` (or `lfilter`) and `prn` parameters of `sniff` to cut down the amount of data that you're processing.

Deliverables:

- A zipped file containing:
  - Your code (.py)
  - A Document explaining which detection methods you decided to implement and how (or if) do you initiate a false positive.
    - If you went on to the additional challenges, add to the document details of your work (including ideas, screen-shots, etc.)

Note:

- You may NOT download or copy code from internet !!
- Searching the internet with specific programming question IS allowed.